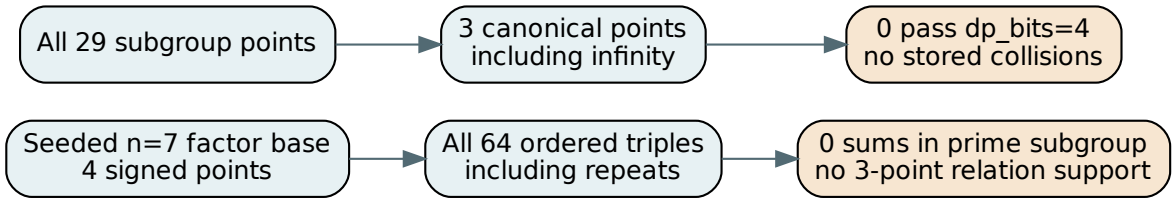
Exact support diagnosis of the n=7 smoke

Post-outcome correctness diagnosis, 2026-10-25, protocol amendment 5. The failed frozen session is ECBS1h1572078f47bd; it exhausted all ten records including both rho controls. This diagnosis changes none of them.

The K0 model has field degree 7, group order 116, prime subgroup r=29 and cofactor 4: icv1-f2m7-t13-616700dd. Its geometric dimension-2 base, seed 17, has four signed points. Exhaustive enumeration of all 64 ordered triples including repeated points finds zero sums in the prime subgroup. Hence none of these three-summand arms can produce a prime-subgroup relation on this exact base; this is a finite support fact, not a failure of F4. Exact identity: `ICV1:f2m-7-e1a6f45a:13:116:0x1:-7:unk:r:616700ddc6fe`; subgroup generator (0x26,0x5).



Enumerating every scalar 0–28 gives three signed-Frobenius canonical representatives including infinity. The low hexadecimal digits of their source-pinned distinguished hashes are 2,3,7. None passes the frozen four-bit distinguished predicate. No canonical point can enter its rho collision table under this configuration. This explains the independent reference failure without treating it as zero rho cost.

Native source: examples/ic_structured_small_audit.rs. Complete coordinates, r-multiples, canonical points and hashes, plus source SHA-256 pins, are in pipeline-receipts/structured-n7-support.json. It reuses actual base construction and public strong-rho canonicalization, and implements the literal private hash from the pinned source. Re-run the native example to check the finite enumeration. No timings are reported; the diagram shows separate failure paths because the two counts have different domains.

This does not establish failure on any other basis or field, retune the original reference, qualify a performance comparison, or prove a generic obstruction. A separately frozen viable tiny fixture is needed for that role.